EEE303 Tutorial 5b

1. A test taken on the 500 MVA alternator yields the following results:
   - Open-circuit line voltage is 15 kV for a DC exciting current of 1400 A.
   - Using the same DC current, with the armature short-circuited the resulting AC line current is 21 000 A.

   Calculate
   a. The base impedance of the generator per phase
   b. The value of the synchronous reactance
   c. The per-unit value of X_s

2. The 3-phase Y-connected generator, 500 MVA, 0.95 power factor, 15 kV, 60 Hz, 200 r/min generator synchronous reactance of 0.4 Ω, per phase. It is connected to an infinite bus having a line voltage of 14 kV, and the excitation voltage is adjusted to 1.14 pu. When the generator delivers total P_e of 420 MW, calculate
   a. The internally induced voltage per phase and terminal voltage per phase
   b. The torque angle
   c. The mechanical displacement angle

3. A 40 MVA, 25 kV, 4-pole, 3-phase generator connected to an infinite bus has a synchronous reactance of 13 Ω per phase. If the internal induced voltage is 14 kV (line-to-neutral), and the system voltage is 21 kV (line-to-line), calculate the following:
   a. The active power which the machine delivers when the mechanical angle is 20°
   b. The peak power that the generator can deliver

4. The synchronous generator has an efficiency of 98.4% when it delivers an output of 500 MW at 200 rpm. Knowing that the DC exciting current is 2400 A at a dc voltage of 300 V, calculate the following:
   a. The total losses in the machine
   b. The copper losses in the rotor
   c. The torque developed by the turbine, consider the copper loss in the rotor even though it is separately excited.

5. Figure T5b-1 shows a per phase equivalent circuit of a generator, and E_o leads E by 30°

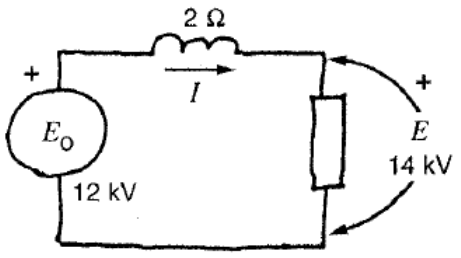


Figure T5b-1

   a. Calculate the total active power output of the generator.
   b. Draw the phasor diagram for one phase.
   c. Calculate the power factor of the load.